

Insights

- There are 177544 rows in the dataset
- First orders were placed on 2015-01-21 09:52:03 and last orders were placed on 2015-02-18 00:30:44
- Majority of the orders are from market id 2
- Market id 6 has the least number of orders
- Order protocol 6 had a slightly higher delivery time
- Majority of the orders are from Order Protocol 1
- Order Protocol 7 has the least number of orders
- From above plot we can see that most of the total items count is between 0 and 15
- At any given time there are around 30-80 outstanding orders
- We can see that during the lunch hours 1-2pm and dinner hours 7-10pm the number of orders is high
- We can see that during the lunch hours 1-2pm and dinner hours 7-10pm the average number of orders is 25 and 45 respectively
- We can see that delivery time increases during peak hours
- We can see that there is a repeating pattern in the daily order data of a month
- We can see that orders are high in first 2 weeks and low in the last week of the month
- Most of the delivery times are under 100 minutes
- Most of the deliveries happen on Friday, Saturday and Sunday
- We can see that American, Pizza, and Mexican are the most ordered categories
- We can see that delivery time decreases as the number of total items increases
- We can see that the delivery time is less as the number of unique items increases

Recommendations

- The company should priorities higher markets that are receiving more orders like market id 2 and deploy more resources to these markets.
- Since Order Protocol 1 has the highest number of orders, the company should focus on improving the delivery time for this protocol.
- Increase the number of delivery partners during peak hours.
- And also during weekends to reduce the delivery time.
- This can be done by hiring more delivery partners or incentivizing existing ones to work during these hours.
- During the start of month, there are higher number of orders, so the company should focus on increasing the number of delivery partners during this time.
- Consider implementing targeted marketing strategies to boost orders in underperforming markets.

Questionnaire

1. Defining the problem statements and where can this and modifications of this be used?
 - The problem statement is to predict the delivery time for a given order based on various features such as order details, restaurant information, and delivery partner availability. This model can be used in various applications such as:
 - E-commerce platforms for estimating delivery times.
 - Food delivery services for providing accurate delivery time estimates to customers.
 - Logistics companies for optimizing delivery routes and times.
 - Ride-hailing services for estimating arrival times of drivers.
2. List 3 functions the pandas datetime provides with one line explanation.
 - `pd.to_datetime()` : Converts a string or number to a datetime object.
 - `pd.date_range()` : Generates a range of dates with specified frequency.
 - `pd.to_timedelta()` : Converts a string or number to a timedelta object.
3. Short note on datetime, timedelta, time span (period)
 - **Datetime**: Represents a specific point in time, including date and time.
 - **Timedelta**: Represents a duration or difference between two datetime objects.
 - **Time Span (Period)**: Represents a span of time, often defined by a start and end date.
4. Why do we need to check for outliers in our data?
 - Outliers can skew the results of statistical analyses and machine learning models, leading to inaccurate predictions and insights. Identifying and handling outliers helps improve model performance and interpretability.
5. Name 3 outlier removal methods?
 - **Z-Score Method**: Identifies outliers by measuring how many standard deviations a data point is from the mean.
 - **IQR Method**: Uses the interquartile range to determine outliers by identifying points that fall below $Q1 - 1.5/IQR$ or above $Q3 + 1.5IQR$.
 - **Isolation Forest**: An algorithm that isolates outliers instead of profiling normal data points.
6. What classical machine learning methods can we use for this problem?
 - **Linear Regression**: A method for modeling the relationship between a dependent variable and one or more independent variables.
 - **LightGBM**: A gradient boosting framework that uses tree-based learning algorithms.
7. Why is scaling required for neural networks?
 - Scaling is required for neural networks to ensure that all input features contribute equally to the model's learning process. It helps in faster convergence during training and prevents certain features from dominating the learning process due to their larger scale.
8. Briefly explain your choice of optimizer.
 - **Adam**: The Adam optimizer is chosen because it combines the advantages of two other extensions of stochastic gradient descent. It computes adaptive learning rates for each parameter and is computationally efficient, requiring little memory. Adam is well-suited for large datasets and high-dimensional spaces, making it a popular choice for training deep learning models.
9. Which activation function did you use and why?
 - **ReLU**: The Rectified Linear Unit activation function is used because it helps mitigate the vanishing gradient problem and allows models to learn faster and perform better.
 - **Elu (After parameter tuning)**: The Exponential Linear Unit activation function is used because it helps to avoid the vanishing gradient problem and has a non-zero mean, which can help with convergence.
10. Why does a neural network perform well on a large dataset?
 - Neural networks can learn complex patterns and relationships in large datasets, allowing them to generalize better and make accurate predictions. The more data available, the better the model can learn and adapt to the underlying distribution of the data.